This PDF was created with https://tianfei.chat , free download: https://tianfei.chat/article/one-click-webpage-to-pdf-tools/?from=saved_pdf_file
From: http://118.185.187.137/moodle/mod/quiz/review.php?attempt=369241&cmid=3325

1-Number of Zeros in a Given ...
2-Majority Element
3-Finding Floor Value
4-Two Elements sum to x
5-Implementation of Quick Sort
Dynamic Programming
1-DP-Playing with Numbers
2-DP-Playing with chessboard
3-DP-Longest Common Subse...
4-DP-Longest non-decreasing ...
Competitive Programming
1-Finding Duplicates-O(n^2) Ti...
2-Finding Duplicates-O(n) Tim...
3-Print Intersection of 2 sorte...
4-Print Intersection of 2 sorte...
5-Pair with Difference-O(n^2)T...

4-DP-Longest non-decreasing Subsequence

Started on	Saturday, 12 September 2026, 2:37 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:46 PM
Time taken	9 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n], dp[n], max_len = 0;
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        dp[i] = 1;
        for (int j = 0; j < i; j++) {
            if (arr[i] >= arr[j] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
            }
        }
        if (dp[i] > max_len) max_len = dp[i];
    }
    printf("%d\n", max_len);
    return 0;
}
```

	Input	Expected	Got	
✔	9 −1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

WEBxPDF
Converting your page to PDF...